COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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Maximum Likelihood Estimation
Part II

Example 1 (Normal Distribution)

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma)$$

Find the MLE of $\boldsymbol{\theta} = (\mu, \sigma^2)$.

Solution:

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} = (2\pi)^{-\frac{1}{2}} (\sigma^2)^{-\frac{1}{2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

$$\Rightarrow L(\mu, \sigma^2) = \prod_{i=1}^{n} \left[(2\pi)^{-\frac{1}{2}} (\sigma^2)^{-\frac{1}{2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}} \right]$$

$$= \prod_{i=1}^{n} (2\pi)^{-\frac{1}{2}} \prod_{i=1}^{n} (\sigma^2)^{-\frac{1}{2}} \prod_{i=1}^{n} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}} = (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} e^{-\sum_{i=1}^{n} \frac{(x_i-\mu)^2}{2\sigma^2}}$$

$$\boxed{\Rightarrow l(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{2\sigma^2}}$$

Example 1 (Solution):

Let $u = \sigma^2$. Then

$$\ell(\mu, u) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log u - \frac{\sum(x_i-\mu)^2}{2u}$$

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2u}\frac{\partial}{\partial \mu}\sum_{i=1}^{n}(x_i-\mu)^2 = -\frac{1}{2u}\sum\frac{\partial}{\partial \mu}(x_i-\mu)^2 = -\frac{1}{2u}\sum 2(x_i-\mu)(-1)$$

$$\frac{\partial \ell}{\partial u} = -\frac{n}{2u} + \frac{\sum(x_i-\mu)^2}{2u^2}$$

Example 1 (Solution):

$$\frac{\partial \ell}{\partial \mu} = \frac{\sum (x_i - \mu)}{u} = 0 \Rightarrow \sum (x_i - \mu) = 0$$

$$\Rightarrow \sum_i^n x_i - \sum_i^n \mu = 0$$

$$\Rightarrow \sum x_i - n\mu = 0$$

$$\Rightarrow \boxed{\hat{\mu}_{MLE} = \frac{\sum x_i}{n}} = \bar{x}$$

$$\frac{\partial \ell}{\partial u} = -\frac{n}{2u} + \frac{\sum (x_i - \mu)^2}{2u^2} = 0$$

$$\Leftrightarrow \left(\frac{\sum (x_i - \mu)^2}{2u^2} = \frac{n}{2u} \right) 2u$$

$$\Leftrightarrow \frac{\sum (x_i - \mu)^2}{u} = n \Leftrightarrow u = \frac{\sum (x_i - \mu)^2}{n} = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$\Rightarrow \hat{\sigma}^2_{MLE} = \frac{\sum (x_i - \bar{x})^2}{n}$$

Example 2

Consider a random sample $X_1, X_2, \dots, X_n$ from the pdf

$$f(x;\theta) = 2x\theta^2, 0 \le x \le \frac{1}{\theta}$$

Find the MLE of θ.

- This is an example where the domain of the distribution depends on the parameter θ.
- In such cases, it helps to write the restrictions with indicator functions:

$$f(x;\theta) = 2x\theta^2 I_{\left[0,\frac{1}{\theta}\right]}(x)$$

where $I_A(x) = \begin{cases} 1, if\ x \in A \\ 0, if\ x \notin A \end{cases}$

Example 2 (Solution):

Consider a random sample $X_1, X_2, \dots, X_n$ from the pdf

$$f(x;\theta) = 2x\theta^2 I_{\left[0,\frac{1}{\theta}\right]}(x)$$

Find the MLE of θ.

Solution: Note that for each observation

$$0 \le x_i \le \frac{1}{\theta} \Leftrightarrow 0 \le \theta \le \frac{1}{x_i}, \forall i = 1, \dots, n$$

so, the indicator can be rewritten as: $I_{\left[0,\frac{1}{\theta}\right]}(x_i) = I_{\left[0,\frac{1}{x_i}\right]}(\theta)$.

Now, the likelihood is:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = 2^n \left(\prod_{i=1}^{n} x_i\right) \theta^{2n} \left(\prod_{i=1}^{n} I_{\left[0,\frac{1}{x_i}\right]}(\theta)\right)$$

Example 2 (Solution):

Aside: Show as exercise that product of indicators is the same as an indicator of intersection:

$$I_A(x)I_B(x) = I_{A\cap B}(x)$$

Therefore,

$$L(\theta) = 2^n\left(\prod_{i=1}^{n} x_i\right)\theta^{2n} I_{\cap_{i=1}^{n}\left[0,\frac{1}{x_i}\right]}(\theta)$$

Finally, notice that

$$I_{\cap_{i=1}^{n}\left[0,\frac{1}{x_i}\right]}(\theta) = I_{\left[0,\,\min\left\{\frac{1}{x_i}\right\}\right]}(\theta) = I_{\left[0,\frac{1}{x_{(n)}}\right]}(\theta)$$

where $x_{(n)} = \max\{x_1, \dots, x_n\}$, so we obtain that

$$L(\theta) = 2^n\left(\prod_{i=1}^{n} x_i\right)\theta^{2n} I_{\left[0,\frac{1}{x_{(n)}}\right]}(\theta)$$

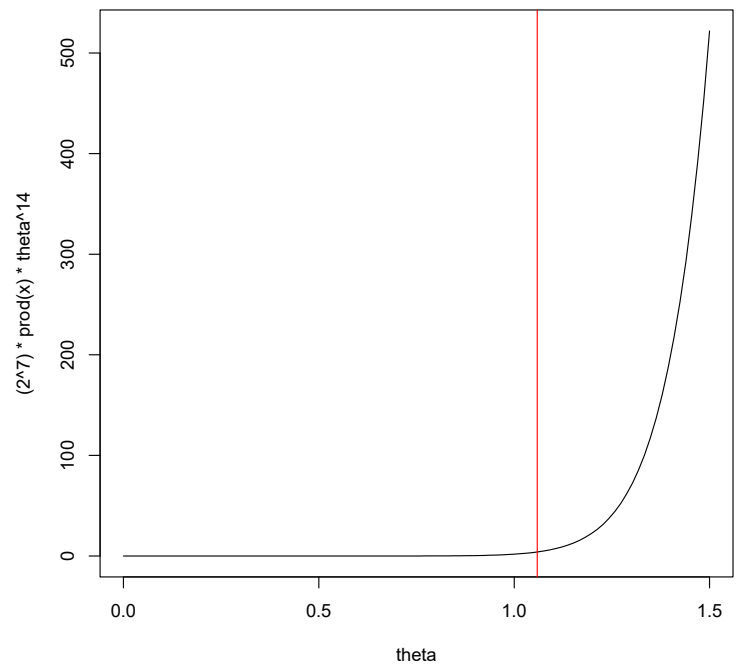
Example 2 (Solution):

The function

$$L(\theta) = 2^n \left(\prod_{i=1}^{n} x_i \right) \theta^{2n} I_{\left[0, \frac{1}{x_{(n)}}\right]}(\theta)$$

is continuous in θ, so by the Extreme Value Theorem, it must have a maximum in its domain, the closed interval $\left[0, \frac{1}{x_{(n)}}\right]$.

However, $L(\theta)$ is strictly increasing, so the max must occur at the right endpoint of the domain (by the definition of increasing functions).



Therefore,

$$\hat{\theta}_{MLE} = \frac{1}{X_{(n)}}$$

Exercise 1 (from STAT 5701)

Let X_1, X_2, ... , X_n be a random sample from the population random variable X with pdf

$$f(x;\theta) = \frac{2x}{1-\theta^2},\ 0 < \theta \leq x \leq 1$$

Show that $\hat{\theta}_{MLE} = \min\{X_1, \dots, X_n\}$.

Exercise 2

Let X_1, X_2, ... , X_n be a random sample from the uniform population random variable $X \sim \mathrm{U}(0, \theta)$.

That is, $f(x;\theta) = \frac{1}{\theta}, 0 \leq x \leq \theta, \theta > 0$

Show that $\hat{\theta}_{\mathrm{MLE}} = \max\{X_1, \ldots, X_n\}$.

Properties of MLEs

Invariance: If $\hat{\theta}_{\mathrm{MLE}}$ is the MLE of θ, then $h(\hat{\theta}_{\mathrm{MLE}})$ is the MLE of $h(\theta)$ for any bijective function h.

This means if we have found already the MLE of a parameter, and we need the MLE of a transformation of the parameter, then we simply transform the already found MLE.

Example 4

Let $X_1, X_2, \ldots, X_n$ be a random sample from the population random variable $X \sim N(\mu, \sigma)$

Find $\hat{\sigma}_{MLE}$.

Recall: $\hat{\sigma}^2_{MLE} = \dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}$ (variance)

$\sigma = \sqrt{\sigma^2}$

$\Rightarrow \hat{\sigma}_{MLE} = \sqrt{\hat{\sigma}^2_{MLE}} = \sqrt{\dfrac{\sum (x_i - \bar{x})^2}{n}}$

Weak Law of Large Numbers

Recall the WLLN:

If $X_1, \ldots, X_n$ is a random sample from *any* distribution with a finite mean μ, then $\bar{X}_n \xrightarrow{P} \mu$ as $n \to \infty$

In many important cases, the sample mean coincides with the MLE of the population mean, so WLLN guarantees that in these cases MLE is consistent!

Q: Is it true in general that any MLE is consistent, even if we are not estimating the mean?

Properties of MLEs

Consistency: If $\hat{\theta}_{\mathrm{MLE}}$ is the MLE of θ, then under some *mild regularity conditions* $\hat{\theta}_{\mathrm{MLE}} \xrightarrow{P} \theta$ as $n \to \infty$.

This means that in most cases the MLEs are guaranteed to be consistent!

Efficiency: Under the same conditions, the MLEs are asymptotically efficient.

Q: What does this even mean?

Sufficient Regularity Conditions

- Identifiability:

$$f(x;\theta_1) \neq f(x;\theta_2) \text{ iff } \theta_1 \neq \theta_2$$

- The support of $f(x;\theta)$ does not depend on θ:

$$\{x: f(x;\theta) > 0\} \text{ does not depend on } \theta$$

- The log-likelihood is differentiable
- The equation $l'(\theta) = 0$ has a unique solution.
- The true value of the parameter is an interior point of the parameter space Θ.

Note: Some of these conditions can be refined and be less restrictive, but they are very technical.

Classic Central Limit Theorem

Recall the CLT: If $X_1, \ldots, X_n$ is a random sample from *any* distribution with standard deviation $\sigma < \infty$, then

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathrm{N}(0,1) \text{ as } n \to \infty$$

The quantity σ^2 is sometimes referred to as the *asymptotic variance* of $\bar{X}_n$, since $\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} \mathrm{N}(0, \sigma)$

Recall that $\bar{X}_n$ is the MLE of the population mean μ.

Q: Can this result be extended to the MLE of any *other parameter*?

Properties of MLEs

Theorem (Asymptotic Normality): If $\hat{\theta}_{\mathrm{MLE}}$ is the MLE of θ, then under some regularity conditions

$$\sqrt{n}\left(\hat{\theta}_{\mathrm{MLE}} - \theta\right) \xrightarrow{d} \mathrm{N}(0, \sigma_{\mathrm{MLE}}) \text{ as } n \to \infty$$

where the asymptotic variance

$$\sigma^2_{\mathrm{MLE}} = \frac{1}{I(\theta)}$$

and $I(\theta) = \mathrm{E}\{[l'(\theta)]^2\} = -\mathrm{E}[l''(\theta)]$ is called the *Fisher information,* which is obtained from a sample of *one observation.*

Equivalently, $I_n(\theta) = n\, I(\theta)$ and $\frac{\hat{\theta}_{\mathrm{MLE}} - \theta}{\sqrt{I_n(\theta)}} \xrightarrow{d} \mathrm{N}(0,1)$

Proofs in Serfling (1980) and Lehmann & Casella (1998)

Example 5:

Find $I_n(\mu)$ for the case of a random sample $X_1, \ldots, X_n \sim N(\mu, \sigma)$ where σ is *known*.

Solution:

$$L(\mu) = (2\pi)^{-\frac{n}{2}}(\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

$$l(\mu) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}$$

$$\Rightarrow l'(\mu) = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma^2}$$

Solution:

$$I_n(\mu) = \frac{n}{\sigma^2} \Leftarrow$$

$$l'(\mu) = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma^2} \Rightarrow l''(\mu) = \frac{-n}{\sigma^2}$$

$$I = E\left(\left[l'(\mu)\right]^2\right) = E\left[\left(\frac{\sum x_i - n\mu}{\sigma^2}\right)^2\right] = \frac{1}{\sigma^4} E\left[\left(\sum x_i - n\mu\right)^2\right]$$

$$= \frac{1}{\sigma^4} Var\left(\sum x_i\right) \quad \text{since } E\left(\sum x_i\right) = n\mu$$

$$= \frac{1}{\sigma^4} n\sigma^2 = \boxed{\frac{n}{\sigma^2}} \Rightarrow \boxed{\bar{X} \approx N\left(\mu, \sqrt{\frac{\sigma^2}{n}}\right)}$$

Example 6 (Bernoulli):

Find $I_1(p)$ of $X \sim$ Bernoulli(p).

Solution:

$$L(p) = p(x) = p^x(1-p)^{1-x}$$
$$\Rightarrow l(p) = \log[p^x(1-p)^{1-x}]$$
$$= x\log p + (1-x)\log(1-p)$$
$$\Rightarrow l'(p) = \frac{x}{p} + (1-x)\frac{1}{1-p}(-1) = \frac{x}{p} - \frac{1-x}{1-p}$$
$$= \frac{X - Xp - p + Xp}{p(1-p)} = \frac{X-p}{p(1-p)} = \frac{X - E(X)}{p(1-p)}$$
$$\Rightarrow I_1(p) = \mathrm{E}\left[\frac{(X-EX)^2}{p^2(1-p)^2}\right] = \frac{V(X)}{p^2(1-p)^2}$$
$$= \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)}$$

Example 7 (Exercise):

Find $I_n(\lambda)$ for the case of a random sample from Poisson(λ).

State the asymptotic normality result.

Efficiency

Cramer-Rao Lower Bound: Let $\hat{\theta}$ be any estimator of θ. Then:

$$Var(\hat{\theta}) \geq \frac{\left[\frac{\partial}{\partial\theta} E(\hat{\theta})\right]^2}{I_n(\theta)}$$

If $\hat{\theta}$ is unbiased the inequality simplifies to

$$Var(\hat{\theta}) \geq \frac{1}{I_n(\theta)}$$

Note: This means that the MLEs are asymptotically efficient. That is, if the sample size increases the MLE achieves the smallest possible variance.

Vector Parameter Space

Suppose we have a vector of parameters $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_m)$. In this case the score function will be an $m \times 1$ *gradient vector*, and the Fisher information will be an $m \times m$ *matrix*.

Definition: Let $X_1, \ldots, X_n$ be a sample with log-likelihood $l(\boldsymbol{\theta})$. The *score vector* is defined as

$$s(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} l(\boldsymbol{\theta}) = \left(\frac{\partial}{\partial \theta_1} l(\boldsymbol{\theta}), \ldots, \frac{\partial}{\partial \theta_m} l(\boldsymbol{\theta}) \right)^T$$

and the *information matrix* is defined as

$$I(\boldsymbol{\theta}) = E[s(\boldsymbol{\theta}) s(\boldsymbol{\theta})^T]$$

Fisher Information Matrix

The *information matrix* is defined as

$$I(\boldsymbol{\theta}) = E[s(\boldsymbol{\theta})s(\boldsymbol{\theta})^T]$$

It can be shown that this means:

$$[I(\boldsymbol{\theta})]_{ij} = E\left[\frac{\partial}{\partial\theta_i} l(\boldsymbol{\theta}) \frac{\partial}{\partial\theta_j} l(\boldsymbol{\theta})\right] = -E\left[\frac{\partial^2}{\partial\theta_i \partial\theta_j} l(\boldsymbol{\theta})\right]$$

Rather than taking reciprocals to obtain an estimate of the variance, we find the matrix inverse $I^{-1}(\boldsymbol{\theta})$. To be more precise, for *n* observations, let $\hat{\theta}_{i,n}$ be the MLE of θi. Then:

$$Var\left(\hat{\theta}_{i,n}\right) \approx \frac{1}{n} I^{-1}(\boldsymbol{\theta})_{ii}$$

Note: The above assumes that the Fisher information matrix was calculated from one observations.

Example 8 (Gamma Distribution):

Let $X_1, \ldots, X_n$ be an iid sample from $\Gamma(\alpha, \lambda)$ distribution
Then $\theta = (\alpha, \lambda)$ and the likelihood function is:

$$L(\alpha, \lambda) = \frac{\lambda^{nr}}{\Gamma(\alpha)^n} \prod_{i=1}^{n} x_i^{\alpha-1} \exp\left(-\lambda \sum_{i=1}^{n} x_i\right)$$

and the log-likelihood is:

$$l(\alpha, \lambda)$$

$$= n\alpha \log(\lambda) - n \log \Gamma(\alpha) + (\alpha - 1) \sum_{i=1}^{n} \log x_i - \lambda \sum_{i=1}^{n} x_i$$

Example 8 (Gamma Distribution):

We need to find the simultaneous solution to:

$$\frac{\partial}{\partial \lambda} l(\lambda, \alpha) = \frac{n\alpha}{\lambda} - \sum_{i=1}^{n} x_i = 0$$

$$\frac{\partial}{\partial \alpha} l(\lambda, \alpha) = n \log(\lambda) - n \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} + \sum_{i=1}^{n} \log(x_i) = 0$$

The first equation implies that:

$$\hat{\lambda} = \frac{\hat{\alpha}}{\bar{x}}$$

and substituting in the second equation we obtain

$$n \log\left(\frac{\hat{\alpha}}{\bar{x}}\right) - n \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})} + \sum_{i=1}^{n} \log(x_i) = 0$$

Example 8 (Gamma Distribution):

We have to solve this equation numerically:

$$n\left(\log(\hat{\alpha}) - \log\bar{x} - \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})}\right) + \sum_{i=1}^{n}\log(x_i) = 0$$

Note: The function $\psi(\hat{\alpha}) = \frac{d}{d\,\hat{\alpha}}\log\Gamma(\hat{\alpha}) = \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})}$ is known as the digamma function and is available in R as `digamma`

Now, let's find the Fisher information matrix.

Recall that:

$$l(\alpha,\lambda)$$

$$= n\alpha\log(\lambda) - n\log\Gamma(\alpha) + (\alpha-1)\sum_{i=1}^{n}\log x_i - \lambda\sum_{i=1}^{n}x_i$$

Example 8 (Gamma Distribution):

$$l(\alpha,\lambda) = n\alpha\log(\lambda) - n\log\Gamma(\alpha) + (\alpha-1)\sum_{i=1}^{n}\log x_i - \lambda\sum_{i=1}^{n} x_i$$

Therefore,

$$[I(\alpha,\lambda)]_{11} = -E\left[\frac{\partial^2}{\partial\alpha^2} l(\alpha,\lambda)\right] = n\frac{d^2}{d\,\alpha^2}\log\Gamma(\alpha)$$

$$[I(\alpha,\lambda)]_{22} = -E\left[\frac{\partial^2}{\partial\lambda^2} l(\alpha,\lambda)\right] = n\frac{\alpha}{\lambda^2}$$

$$[I(\alpha,\lambda)]_{12} = -E\left[\frac{\partial^2}{\partial\alpha\partial\lambda} l(\alpha,\lambda)\right] = -n\frac{1}{\lambda}$$

Note: The second derivative of $\ln\Gamma(\alpha)$ is known as the trigamma function and is denoted $\psi_1(\alpha)$ or `trigamma` in `R`

Example 8 (Gamma Distribution):

This gives us the following Fisher information matrix:

$$I(\alpha,\lambda) = n\begin{bmatrix} \psi_1(\alpha) & -\dfrac{1}{\lambda} \\ -\dfrac{1}{\lambda} & \dfrac{\alpha}{\lambda^2} \end{bmatrix}$$

The above result can be used to estimate the variances of the MLEs of α and λ by finding $I^{-1}(\alpha,\lambda)$ and plugging in the MLE values in place of α and λ.

Note: You can obtain the MM estimators of α and λ much easier and you can derive their variances easier, but the MM variances will be higher!

Exercise (Weibull Distribution):

Let $X_1, \ldots, X_n$ be a random sample from the Weibull distribution with pdf

$$\frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-\left(\frac{x}{\lambda}\right)^k}, x \geq 0$$

where $k > 0, \lambda > 0$ are the unknown parameters.
It is used to model time to "failure", like time user spends on a webpage.
Then we have a vector $\boldsymbol{\theta} = (k, \lambda)$

Find the likelihood function and try to maximize it.